

Hackathon Problem Statements

Regd No: 25091A32P2, 25091A32P4

Problem Statement No: 80

Rainwater Harvesting Potential Estimator Design a rainwater harvesting potential calculator using regression on rooftop area and regional rainfall datasets to estimate annual harvestable water volume for individual buildings or entire neighborhoods. Use OOP to model buildings, catchment areas, and harvesting systems. This tool can support municipal rainwater harvesting mandate compliance and promote household-level water conservation.

Regd No: 25091A32C4, 25091A32C8

Problem Statement No: 94

Digital Wallet Transaction Routing Optimizer Create a transaction routing optimization system using graph-based network algorithms to select the fastest, lowest-cost path for digital payment transactions across multiple banking/payment gateway nodes. Use OOP to model payment gateways, transaction routes, and cost/latency metrics. This addresses the reliability and speed challenges faced by digital payment systems during high-load periods like festival sales.

Regd No: 25091A32I5, 25091A32H6

Problem Statement No: 110

Cyclone Path Prediction Visualizer Build a cyclone trajectory visualization tool using regression analysis on historical cyclone path datasets to project likely future paths and affected coastal regions based on current position and speed data. Use OOP to model cyclone tracking data and affected-zone predictions. This supports coastal disaster management authorities in issuing timely, region-specific evacuation advisories.

Regd No: 25091A32K5, 25091A3241

Problem Statement No: 125

Unemployment Trend Analyzer Build an unemployment trend analysis dashboard using time-series analysis on regional job statistics datasets to identify sectors and districts with rising unemployment, supporting targeted skill-development and job-creation policy planning. Use OOP to structure data sources and regional analysis modules. This gives policymakers timely, granular insight beyond aggregate national unemployment figures.

Regd No: 25091A32C7, 25091A32J1

Problem Statement No: 140

Digital Identity Duplicate-Detection System Build a duplicate identity detection system using string similarity algorithms (edit distance, phonetic matching) to identify likely duplicate or fraudulent entries in a beneficiary database, even when names are misspelled or formatted differently. Use OOP to model identity records and match-confidence scores. This helps prevent duplicate benefit claims across large government welfare databases.

Regd No: 25091A3292, 25091A3291

Problem Statement No: 146

Social Media Bot Account Detector Design a bot account detection tool using classification on behavioral pattern features (posting frequency, account age, follower-following ratio, content repetition) from public social media datasets to flag likely automated accounts. Use OOP to model accounts, behavior metrics, and detection scores. This helps address the growing problem of coordinated inauthentic behavior and misinformation amplification on social platforms.